

14. Analysis E: boundary detection bias

At a boundary, the sensor can err in two directions. It can report a value where geometry says no stereo reading is possible (a fabricated read, usually interpolated). Or it can report a no-read where a valid surface was visible (a dropout). Comparing the two rates shows whether the VSX3000 leans toward reading values or toward no-reads. A second measure shows which surface it favors when it does read: the near (front) surface or the far (back) one. Both come from the B edge data and the C array data, so no new captures are needed.

1. Geometric visibility mask. For every pixel in the 8 px boundary band ($|s|$ at most 8 px, `BOUNDARY_BAND_HALF_WIDTH_PX`), ray-cast its true surface point from the registered pose. Test whether that point is seen by the left camera and the right camera and lit by the projector at its as-built position (`PROJECTOR_OFFSET_MM`). Set $V = 1$ if a depth read is geometrically possible and $V = 0$ if not. As in Section 12, Step 7, compute V both with and without the projector condition: the depth-read rule with the projector is the primary one, the two-camera rule the check, and the two coincide for a convex outline when the projector is midway between the cameras.
2. Outcome classes. Classify each pixel-frame in the band as one of four outcomes. Front read: valid and within $3 \times \sigma_{\text{tot}}(Z)$ of the front plane (`SURFACE_ASSIGNMENT_SIGMA_MULTIPLE`). Back read: the same test against the back plane. Intermediate read: valid but near neither plane (a "flying pixel"). No-read.
3. Outcome profiles. Bin by signed distance s to the true edge in 0.25 px bins (`BOUNDARY_BIN_WIDTH_PX`), with $s > 0$ on the front side. Plot $P_{\text{front}}(s)$, $P_{\text{back}}(s)$, $P_{\text{mid}}(s)$, and $P_{\text{none}}(s)$, which sum to 1. Use radial distance $r - D/2$ for disks and cutouts. Pool over phase-jitter poses so every sub-pixel phase is equally represented.
4. Read vs no-read bias. Compute the fabrication rate R_{fab} and the dropout rate R_{drop} as below. Combine them into a bias index β_{read} from -1 (always drops out) to $+1$ (always fills in). Also report two equivalent widths in px: W_{fab} , how far reads extend into the geometrically unreadable region, taken as the sum over the bins with $V = 0$ of the fraction of pixels read in the bin times the bin width; and W_{drop} , how far no-reads intrude into the readable region, the same sum over the bins with $V = 1$ of the fraction of no-reads. Widths are easier to use in a forward model than rates.

$$R_{\text{fab}} = P(\text{read}|V=0), \quad R_{\text{drop}} = P(\text{no-read}|V=1), \quad \beta_{\text{read}} = \frac{R_{\text{fab}} - R_{\text{drop}}}{R_{\text{fab}} + R_{\text{drop}}}$$

$$W_{\text{fab}} = \int_{V=0} P_{\text{read}}(s) ds, \quad W_{\text{drop}} = \int_{V=1, |s| \leq s_{\text{band}}} P_{\text{none}}(s) ds$$

5. Near vs far preference. Count wrong-surface reads: front reads at $s < 0$ (N_{fb}) and back reads at $s > 0$ (N_{bf}). The index $\pi_{\text{near}} = (N_{\text{fb}} - N_{\text{bf}})/(N_{\text{fb}} + N_{\text{bf}})$ runs from -1 (favors far) to $+1$ (favors near). Stereo matchers with a correlation window usually show positive π_{near} ("foreground fattening"), but the VSX3000's behavior is not known in advance. Cross-check against the edge offset s_{50} from 11.1, Step 8, and the sign of b from Section 12: with their conventions, foreground fattening means $\pi_{\text{near}} > 0$, $s_{50} < 0$, and $b > 0$ for disks and $b < 0$ for cutouts. Also report the mean h of intermediate reads: 0.5 means even blending, and values near 0 or 1 mean blending that leans to one surface.
6. Feature-scale bias. For each cutout diameter, classify the region inside the true outline by majority outcome in each frame: back read (correct), front read (fill-in), or no-read. Plot $P(\text{fill-in})$, $P(\text{no-read})$, and $P(\text{correct})$ against D_{px} . The fill-in fraction near D_{50} shows directly whether the sensor reports a solid surface or a hole where it cannot resolve. For disks, the matching measure is the fraction of disks reported as back plate (erased) versus no-read. Also report the difference between the two detection rules of Section 13, Step 2, at each D . The classification uses the D trials (one frame per pose) and the C frames, so the fill-in fraction near D_{50} rests on the same trial counts as Section 13.
7. Breakdowns. Report β_{read} , W_{fab} , W_{drop} , π_{near} , and s_{50} for each edge orientation (along the baseline vs across it), polarity, Z , and G , and for straight edges vs circular features, under both visibility rules. Confidence intervals for β_{read} , π_{near} , W_{fab} , and W_{drop} come from bootstrapping over poses with 2000 resamples (BOOTSTRAP_RESAMPLES).
8. Outputs. `E_boundary_bias.csv` with one row per breakdown of Step 7 and per visibility rule, and `E_boundary_details.json`. Figures (PNG and SVG): stacked outcome profiles against s for each edge type; β_{read} and π_{near} against Z ; feature-scale outcome fractions against D_{px} . W_{fab} , W_{drop} , π_{near} , and β_{read} go in `forward_model_parameters.json` as the boundary terms of the forward model.